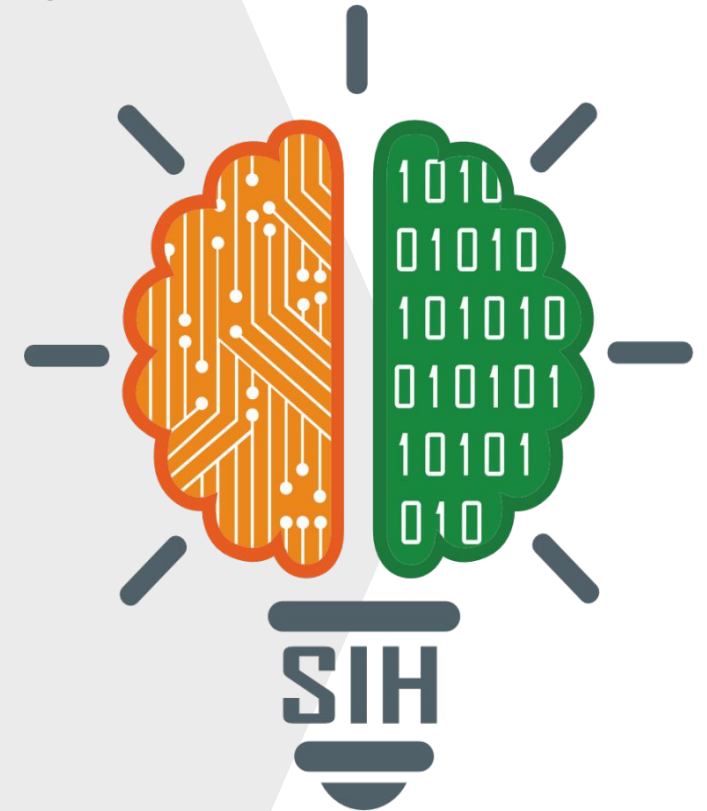




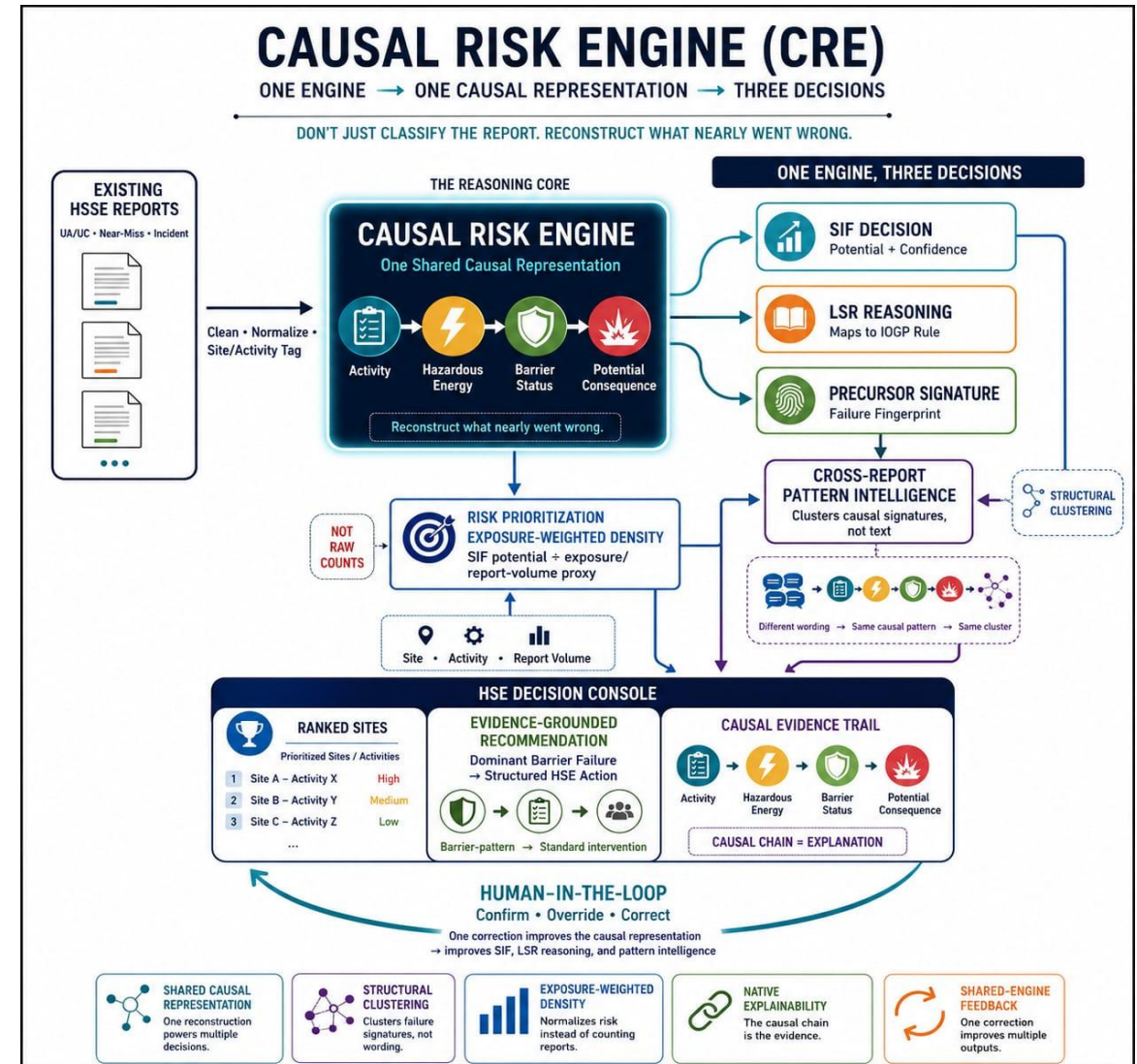
PRAVAH : Causal Risk Intelligence Engine

- **Problem Statement ID** – SIH26165
- **Problem Statement Title-** AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** **NIL**
- **Team Name-** Nexora



Proposed Solution :

- **Reconstructs the Risk, Not Just the Text**
Activity → Hazardous Energy → Barrier Failure → Potential Consequence
- **One Causal Engine. Three Critical Decisions.**
SIF Potential • Life-Saving Rule • Recurring Precursor Pattern
- **Finds the Same Failure Behind Different Words**
Connects differently worded reports to the same **underlying failure mode**.
- **Prioritizes Where HSE Action Matters Most**
Exposure-weighted SIF density reveals real **fatal-potential risk** — not just high report volume.
- **Gets Smarter With Every Expert Review**
Human-in-the-loop feedback continuously calibrates the system for **real-world deployment**.
- **Built for OIL's Existing Safety Workflow**
Works directly on **UA/UC observations, near-miss** and incident reports — without creating a parallel reporting process.



Technology Stacks

Frontend: Streamlit

Backend: Python, FastAPI, Pydantic, Uvicorn

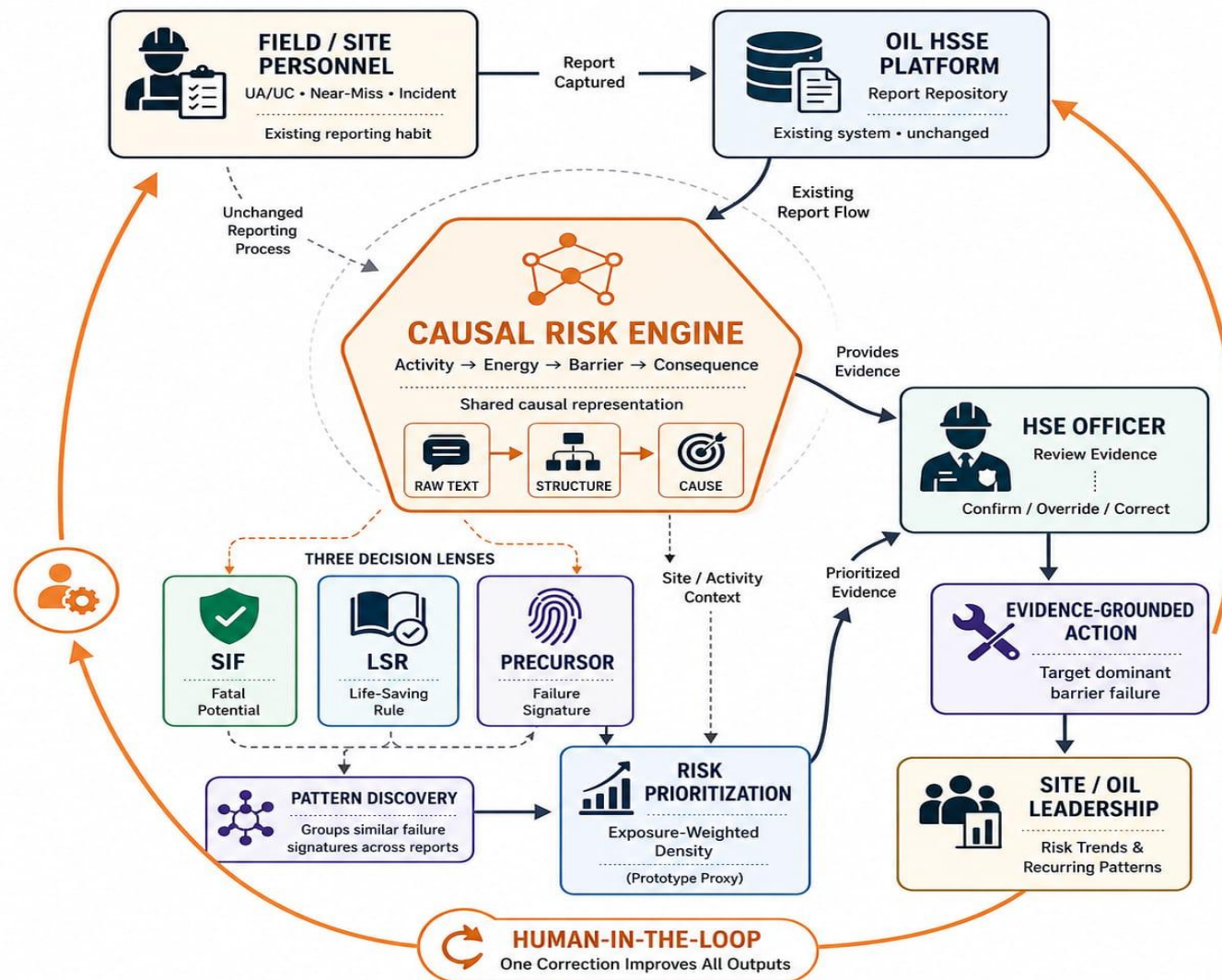
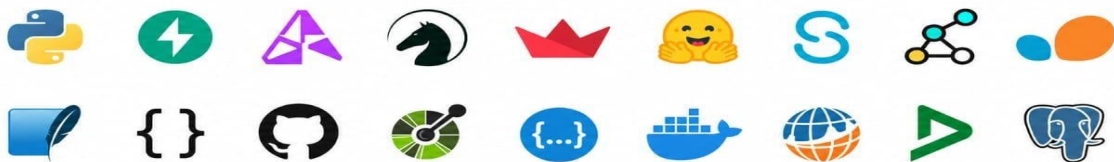
AI/ML & NLP: Hugging Face Transformers, spaCy,
Sentence-Transformers, scikit-learn

Causal Reasoning: Rule Engine, Deterministic Scoring, SIF/LSR
Logic

Data & Knowledge: SQLite, JSON, IOGP Life-Saving Rules,
DEKRA/EEI Taxonomy

API & Services: REST API, OpenAPI

Cloud & Deployment: GitHub, Local Deployment, Streamlit
Community Cloud



Feasibility

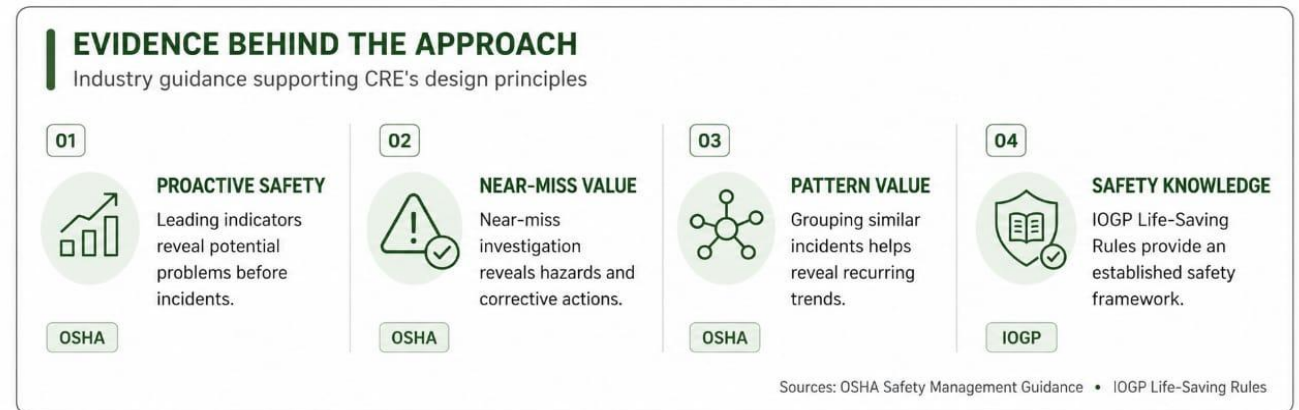
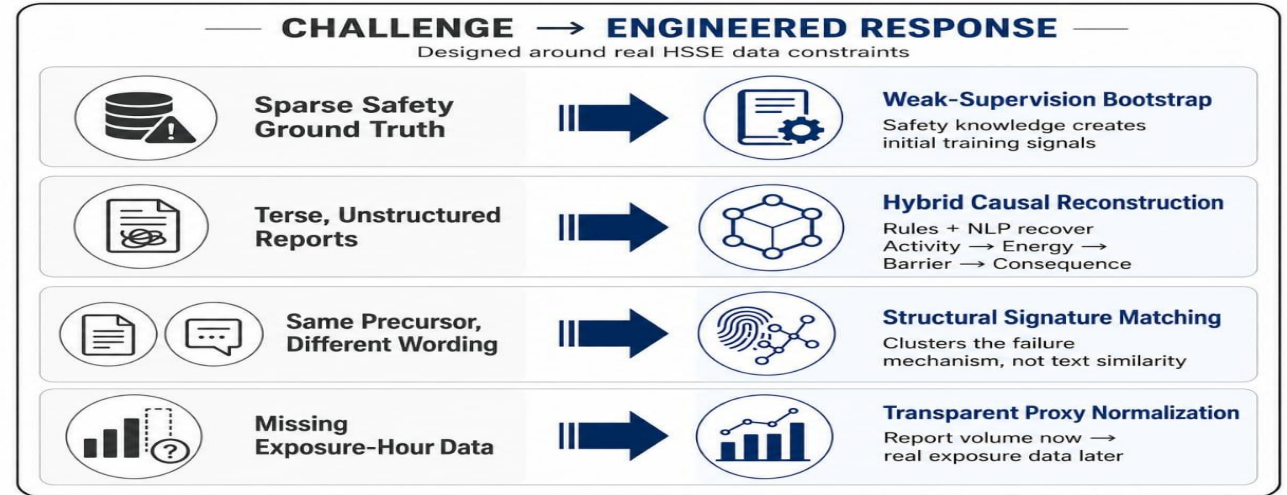
1. **One Core:** Three decisions, one causal structure
2. **No New Data:** Existing HSSE reports are enough
3. **No Paid AI :** Open-source core, optional APIs
4. **Built to Scale :** Site → enterprise rollout

Viability

1. **From Reports to Risk :** Prioritize before escalation
2. **Same Failure, Different Words :** Structure, not text
3. **No Workflow Disruption :** Existing reporting stays unchanged
4. **One Correction, Three Improvements:** Shared-engine feedback

Use Case

1. **Hidden SIFs :** Potential beyond reported severity
2. **Recurring Precursors :** Invisible patterns surfaced
3. **Risk Hotspots :** Act where risk concentrates
4. **Barrier Failures :** Rule-linked intervention





Potential impact on the target audience:

- **Field Personnel** : Existing reporting stays unchanged while earlier risk signals trigger faster preventive action.
- **HSE Officers** : Causal reconstruction converts scattered reports into prioritized, evidence-backed interventions.
- **Site Leadership** : Recurring barrier failures become visible before they escalate.
- **OIL Management** : Shared risk intelligence enables consistent safety prioritization across sites.
- **HSE Decision-Makers** : Every flag carries causal evidence, making AI outputs easier to validate and act upon.

Benefits of the solution

SAFETY

- Detects hidden high-consequence potential
- Exposes recurring barrier failures
- Strengthens preventive safety decisions

OPERATIONAL

- Prioritizes where HSE acts first
- Cuts manual report screening
- Accelerates evidence-based intervention

ECONOMIC

- Prevents costly recurring failures
- Targets resources where risk concentrates
- Reduces avoidable incident exposure

TECHNOLOGICAL

- Converts free text into causal intelligence
- Unifies SIF, LSR & precursor decisions
- Learns from human corrections

ENVIRONMENTAL

- Surfaces precursors to environmental harm
- Enables earlier preventive intervention
- Supports proactive environmental risk control

REFERENCES

INDUSTRY SAFETY FRAMEWORKS

[IOGP — Life-Saving Rules](#)

Established framework for critical hazard/control mapping

[OSHA — Leading Indicators](#)

Proactive indicators for identifying potential problems early

RESEARCH & BEST PRACTICES

[Constructs of Leading Indicators: A Synthesis of Safety Literature](#) - Research foundation for identifying safety precursors and enabling continuous improvement.

[A Scoping Literature Review of NLP for Application to Safety Occurrence Reports](#) - NLP-based analysis of free-text safety reports

INDUSTRY BASED PRACTICES

[DEKRA — Leading Indicators](#)

Proactive measurement for early safety improvement.

[NLP for Industrial Safety Reports.](#)

AI-based analysis of safety-report patterns.

